oceans. Instead, Venus is a stifling, brooding inferno. There are some deserts, but it's mainly a world of frozen lava seas. Our hopes are unfulfilled. The call of this world is now more muted than in the early days of spacecraft exploration, when almost anything was possible and our most romantic notions about Venus might, for all we then knew, be realized.

MANY SPACECRAFT CONTRIBUTED to our present understanding of Venus. But the pioneering mission was *Mariner 2*. *Mariner 1* failed at launch and —as they say of a racehorse with a broken leg—had to be destroyed. *Mariner 2* worked beautifully and provided the key early radio data on the climate of Venus. It made infrared observations of the properties of the clouds. On its way from Earth to Venus, it discovered and measured the solar wind—the stream of charged particles that flows outward from the Sun, filling the magnetospheres of any planets in its way, blowing back the tails of comets, and establishing the distant heliopause. *Mariner 2* was the first successful planetary probe, the ship that ushered in the age of planetary exploration.

It's still in orbit around the Sun, every few hundred days still approaching, more or less tangentially, the orbit of Venus. Each time that happens, Venus isn't there. But if we wait long enough, Venus will one day be nearby and *Mariner 2* will be accelerated by the planet's gravity into some quite different orbit. Ultimately, *Mariner* 2, like some planetesimal from ages past, will be swept up by another planet, fall into the Sun, or be ejected from the Solar System.

Until then, this harbinger of the age of planetary exploration, this minuscule artificial planet, will continue silently orbiting the Sun. It's a little as if Columbus's flagship, the *Santa María*, were still making regular runs with a ghostly crew across the Atlantic between Cádiz and Hispaniola. In the vacuum of interplanetary space, *Mariner 2* should be in mint condition for many generations.

My wish on the evening and morning star is this: that late in the twenty-first century some great ship, on its regular gravity-assisted